

3.0 Milestone Assessment: Network Topologies

Enterprise Network Redesign

In this walkthrough, I redesign a growing company's network after expansion into a second building. The goal is to improve performance, reliability, scalability, and security using structured network engineering practices.

Overview

The existing network exhibited several critical issues. Network slowdowns were caused by improper Virtual Local Area Network (VLAN) configuration, and the absence of redundancy resulted in outages when links failed. Wireless coverage was inconsistent, with dead zones in key areas such as conference rooms. Additionally, there was no separation between internal and guest wireless traffic, creating potential security risks.

This walkthrough demonstrates how to analyze these problems and implement a structured, scalable solution using industry-standard networking technologies.

Objectives

The objectives of this redesign are to:

- Implement proper VLAN segmentation to improve performance and security
- Configure trunking to support communication between VLANs
- Introduce redundancy to eliminate single points of failure
- Prevent switching loops using Spanning Tree Protocol (STP)
- Deploy a reliable and scalable wireless network
- Separate and secure internal and guest network access

Step 1: Assess the Physical Network

Before making configuration changes, the physical network must be evaluated to ensure the design is feasible and scalable. This includes verifying fiber connectivity between buildings to support high-speed, long-distance communication and confirming that copper Ethernet runs do not exceed the 100-meter limitation. High-density areas such as conference rooms and shared workspaces must be identified, as these locations require increased network capacity and wireless coverage.

It is also important to estimate the number and types of connected devices, including workstations, Voice over IP (VoIP) phones, wireless devices, and Internet of Things (IoT) devices. This information

ensures the network can support current demand and future growth. A thorough physical assessment prevents design limitations and reduces the need for future reconfiguration.

Step 2: Design VLAN Segmentation

The original network used a flat design in which all devices were placed on the same subnet. This created excessive broadcast traffic, reduced performance, and introduced significant security risks. To resolve these issues, the network will be segmented into multiple VLANs based on department and function. For example, separate VLANs will be created for Human Resources, Finance, Sales, Information Technology, servers, voice traffic, guest wireless access, and network management.

Segmenting the network into VLANs reduces broadcast domains, improves performance, and enhances security by limiting communication between unrelated devices. For instance, guest devices placed in a dedicated VLAN will be isolated from internal systems and prevented from accessing sensitive resources. This segmentation also simplifies troubleshooting and enables better control of network policies.

VLAN Design

Department	VLAN
HR	VLAN 10
Finance	VLAN 20
Sales	VLAN 30
IT	VLAN 40
Servers	VLAN 50
Voice	VLAN 60
Guest WiFi	VLAN 70
Management	VLAN 99

Step 3: Configure Trunk Links and Inter-Switch Connectivity

To allow communication between VLANs across multiple switches and buildings, trunk links must be configured. These links use IEEE 802.1Q to carry traffic from multiple VLANs over a single physical connection. Fiber optic uplinks will be used between buildings to provide high bandwidth and support

longer distances without signal degradation. Trunking ensures that all VLANs are consistently available throughout the network, enabling seamless communication between devices located in different areas.

Step 4: Add Redundancy and High Availability

The original network lacked redundancy, creating single points of failure. To improve reliability, redundant uplinks will be added between core and distribution switches, providing multiple paths for network traffic.

In addition, link aggregation will be implemented using the Link Aggregation Control Protocol (LACP). This protocol combines multiple physical links into a single logical connection, increasing bandwidth while also providing failover capability if one link fails. These improvements ensure continuous network operation and significantly reduce the risk of outages.

Step 5: Prevent Switching Loops

Redundant network links improve availability but can introduce switching loops if not properly controlled. These loops can severely impact network performance by causing broadcast storms, instability in Media Access Control (MAC) address tables, and significant network slowdowns.

To prevent these issues, the Rapid Spanning Tree Protocol will be implemented. RSTP dynamically blocks redundant paths until they are needed, ensuring loop-free operation while still allowing rapid failover in the event of a link failure. The core switch will be configured as the root bridge to maintain predictable and optimized traffic flow throughout the network.

Step 6: Implement Layer 3 Switching

To improve routing efficiency, Layer 3 switches will be used instead of relying solely on traditional routers. Layer 3 switching enables inter-VLAN routing, which allows communication between different VLANs directly within the switching infrastructure.

This approach reduces latency and improves overall network performance by handling routing decisions closer to the source of traffic. For example, if the Finance VLAN requires access to server resources, the Layer 3 switch can route that traffic efficiently without sending it through an external router.

Step 7: Redesign Wireless Network

The existing wireless infrastructure resulted in inconsistent coverage, including dead zones and unreliable connectivity in high-traffic areas. To resolve these issues, a redesigned wireless network will be implemented using enterprise-grade technologies and strategic planning.

Wireless Standards and Performance

To ensure scalability and high performance, modern wireless technologies such as Wi-Fi 6 and Wi-Fi 6E will be deployed. These standards support both the 5 GHz and 2.4 GHz frequency bands, allowing high-speed connectivity for modern devices while maintaining compatibility with legacy equipment. These improvements provide better performance in high-density environments, increase the number of supported devices, and enable smoother roaming between access points.

Step 8: Wireless Network Segmentation and Security

To ensure both security and performance, the wireless network will be segmented into separate Service Set Identifiers (SSIDs) for corporate and guest access.

Corporate Wireless Network

The corporate wireless network will use the SSID Company-Internal and will be secured using WPA3 Enterprise. Authentication will be managed through a Remote Authentication Dial-In User Service (RADIUS) server, which provides centralized authentication, authorization, and accounting.

Each user will be assigned unique credentials, and integration with directory services will ensure that access is controlled based on identity. This configuration provides strong authentication and secure access for authorized employees.

Guest Wireless Network

A separate guest network will be configured using the SSID Company-Guest and assigned to a dedicated VLAN to isolate guest traffic from the internal network. Access to this network will be restricted to internet-only connectivity, preventing access to internal resources. Bandwidth limitations may also be applied to ensure fair usage among guests. Optionally, a Captive portal can be implemented to require users to accept terms of service before accessing the network.

Step 9: Network Monitoring and Additional Security Hardening

Centralized logging and monitoring will be implemented using Syslog and network monitoring dashboards to provide continuous visibility into network performance and security events. Switch-level hardening will include enabling port security and administratively disabling unused ports to prevent unauthorized access. Additionally, network monitoring will be performed using Simple Network Management Protocol to track device status and performance metrics.

As an additional layer of control, Media Access Control (MAC) address filtering may be implemented to restrict network access to known devices; however, this method is not solely relied upon due to its susceptibility to spoofing. Instead, it is incorporated as part of a layered, defense-in-depth security

approach. Together, these controls reduce the overall attack surface by limiting potential entry points, enhance network visibility, and enable rapid detection and response to anomalies. For stronger authentication, technologies such as IEEE 802.1X could be implemented in future designs to provide identity-based network access control.

Step 10: Validation and Testing

After implementation, the network must be tested to ensure that all improvements are functioning as intended. This includes verifying failover by disconnecting redundant links, confirming VLAN isolation between departments, validating guest network restrictions, and testing wireless roaming between buildings. Performance metrics should also be measured to ensure that latency and throughput have improved. Thorough testing ensures the network is stable, secure, and ready for production use.

Key Takeaways

This redesign demonstrates how proper network segmentation, redundancy, and wireless planning can significantly improve performance, reliability, and security. Implementing VLANs reduces broadcast traffic and enhances isolation, while redundancy and RSTP ensure high availability and stability. Layer 3 switching improves efficiency, and enterprise wireless design ensures consistent and secure connectivity. These principles form the foundation of both modern networking and cybersecurity practices.

This project reflects a structured and scalable approach to network design. By addressing both wired and wireless challenges, the resulting network is more resilient, secure, and capable of supporting future growth. The integration of monitoring, segmentation, and layered security controls ensures that the network can be effectively managed and protected over time.